

PRAIRIE RIVER, AB, Canada

WMO: 715930

Lat: 58.627N

Lon: 111.679W

Elev: 695

StdP: 14.33

Time Zone: -7.00 (NAM)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-27.7	-40.4	0.6	-32.8	-34.6	0.8	-27.5	22.9	4.4	20.4	4.0	5.3	290	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	80.4	63.1	77.5	62.2	74.4	61.0	66.0	76.8	64.4	73.8	62.7	71.1	8.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	85.0	70.5	60.3	80.2	68.8	58.9	76.2	67.2	31.1	76.8	29.8	74.1	28.6	71.4	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.3	16.2	DB	-38.4	87.0	4.9	2.6	-41.9	88.9	-44.8	90.4	-47.5	91.9	-51.0	93.8	(4)
(5)			WB	-38.6	70.0	4.9	2.6	-42.1	71.9	-44.9	73.5	-47.7	74.9	-51.2	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.2	-1.4	0.4	11.5	31.7	45.7	57.8	63.0	58.3	48.9	35.3	17.6	3.5		
	DBStd	25.57	16.51	13.27	14.72	12.00	9.87	7.18	5.23	6.84	7.63	9.12	11.68	14.46		
	HDD50	7939	1593	1389	1193	553	194	17	0	12	112	464	971	1441		
	HDD65	12416	2058	1809	1658	999	601	232	100	225	485	922	1421	1906		
	CDD50	1072	0	0	0	4	62	251	402	268	78	7	0	0		
	CDD65	73	0	0	0	0	4	16	37	16	0	0	0	0		
	CDH74	718	0	0	0	0	48	189	322	150	8	1	0	0		
	CDH80	96	0	0	0	0	5	34	34	23	0	0	0	0		
	WSAvg	7.9	7.6	7.8	8.5	8.5	9.1	8.2	7.4	7.4	7.8	7.9	7.7	7.3		
	PrecAvg	14.8	0.6	0.6	0.6	0.6	1.1	2.1	2.6	2.1	2.4	0.8	0.8	0.7		
(15) Precipitation	PrecMax	19.3	1.3	1.7	1.4	2.5	2.3	4.8	4.3	4.7	5.8	2.4	2.3	2.3		
	PrecMin	10.8	0.2	0.2	0.1	0.1	0.4	0.7	0.9	0.7	0.8	0.0	0.2	0.2		
	PrecStd	2.3	0.3	0.3	0.4	0.5	0.6	1.0	1.0	1.2	1.3	0.6	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	37.2	48.0	66.1	79.9	84.5	83.6	82.9	74.6	65.2	43.9	40.6		
		MCWB	36.3	32.6	39.4	49.8	57.9	63.3	64.5	67.2	61.5	54.5	38.0	36.6		
	2%	DB	33.7	30.3	41.9	59.3	72.9	79.1	80.4	78.2	68.6	57.3	39.2	33.8		
		MCWB	30.6	27.7	35.6	45.4	54.2	59.1	63.9	65.5	58.4	48.2	34.8	31.6		
	5%	DB	28.4	25.0	38.0	53.9	66.0	75.3	77.8	74.2	64.6	51.8	35.4	29.2		
		MCWB	26.8	22.8	33.5	43.2	51.3	58.5	63.3	62.9	55.2	44.6	33.0	27.8		
	10%	DB	22.0	19.8	33.8	48.5	61.6	71.4	74.8	70.9	61.0	48.2	32.8	25.2		
		MCWB	21.0	18.7	31.0	40.5	49.0	57.3	62.3	61.1	53.9	42.4	31.4	24.1		
	0.4%	WB	35.9	33.3	39.6	50.9	59.6	66.1	68.3	69.2	62.8	54.8	38.4	36.4		
		MCDB	38.8	37.2	47.1	64.6	76.4	78.7	78.4	79.3	71.5	65.5	42.9	40.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	30.8	27.7	36.0	46.7	55.4	63.1	66.2	66.1	59.4	49.1	35.2	32.0		
		MCDB	33.5	30.3	41.2	57.0	67.9	72.6	76.8	76.7	66.9	53.8	37.9	33.5		
	5%	WB	26.7	22.9	33.7	43.7	52.6	60.6	64.8	63.8	56.8	45.3	33.2	27.8		
		MCDB	28.6	24.9	37.7	53.0	64.5	69.9	74.4	72.4	62.6	51.5	35.2	29.1		
	10%	WB	21.0	18.7	30.6	40.8	50.0	58.8	63.4	61.9	54.6	42.7	31.4	24.1		
		MCDB	21.9	19.8	34.1	49.0	59.0	69.4	72.4	69.4	60.0	47.6	32.9	25.1		
	5% DB	MDBR	15.8	19.4	21.6	21.1	19.9	21.1	19.8	18.8	17.1	15.0	12.7	14.9		
		MCDBR	19.0	24.6	24.7	27.1	26.2	26.2	24.4	24.3	22.6	21.5	14.7	17.6		
		MCWBR	16.7	22.2	19.2	15.7	12.3	11.9	12.1	13.4	13.2	13.0	11.6	15.6		
		MCWBR	19.2	24.3	23.4	25.9	23.3	21.5	21.9	23.0	20.1	20.2	14.0	17.2		
(40) Clear-Sky Solar Irradiance	taub		0.219	0.244	0.284	0.327	0.348	0.370	0.391	0.355	0.318	0.286	0.234	0.197		
	taud		2.215	2.307	2.273	2.362	2.412	2.351	2.360	2.435	2.542	2.545	2.294	2.117		
	Ebn at Noon		207	259	274	280	281	274	265	268	264	242	198	173		
	Edh at Noon		14	21	30	32	33	36	35	30	23	17	13	10		
(44) All-Sky Solar Radiation	RadAvg		157	436	951	1510	1834	1902	1758	1420	921	444	170	95		
	RadStd		17	32	52	93	155	132	102	79	79	52	21	8		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page